CompoundIQ — API Reference

Version: 1.0 | **Date:** April 5, 2026 **Base URL:** `https://functional-medicine-infrastructure.vercel.app/api`

Overview

CompoundIQ exposes a RESTful API built on Next.js App Router Route Handlers, deployed as Vercel serverless functions. All endpoints use JSON request/response bodies, return standard HTTP status codes, and follow consistent error envelope formatting.

Authentication: Most endpoints require a Supabase Auth JWT (passed as a session cookie). Webhook endpoints use service-specific signature verification. Cron endpoints use CRON_SECRET bearer token.

Error Response Format

All errors return a consistent JSON envelope:

```
{
  "error": {
    "code": "STATE_TRANSITION_INVALID",
    "message": "Order is not in the expected state",
    "details": "Expected AWAITING_PAYMENT, found PAID_PROCESSING",
    "request_id": "req_abc123",
    "timestamp": "2026-04-05T14:30:00.000Z"
  }
}
```

Error Codes

Code	HTTP Status	Description
ORDER_NOT_FOUND	404	Order ID does not exist
STATE_TRANSITION_INVALID	409	CAS failed — order not in expected state
PATIENT_NOT_FOUND	404	Patient record not found
MEDICATION_NOT_FOUND	404	Medication not in catalog
PHARMACY_NOT_FOUND	404	Pharmacy not configured
PAYMENT_REQUIRED	402	Payment has not been completed
CHECKOUT_EXPIRED	410	Checkout JWT token has expired (72-hour window)
FAX_SEND_FAILED	502	Documo fax dispatch failed
SMS_SEND_FAILED	502	Twilio SMS dispatch failed
WEBHOOK_SIGNATURE_INVALID	401	Webhook signature verification failed

RATE_LIMITED	429	Too many requests — check Retry-After header
IDEMPOTENCY_CONFLICT	409	Duplicate request detected
CIRCUIT_BREAKER_OPEN	503	Pharmacy integration temporarily unavailable
ADAPTER_SUBMISSION_FAILED	502	Order submission to pharmacy failed
ADAPTER_TIER_EXHAUSTED	502	All pharmacy channels exhausted
AI_CONFIDENCE_BELOW_THRESHOLD	422	Tier 2 portal submission needs manual review
PHARMACY_WEBHOOK_INVALID_SIGNATURE	401	Pharmacy webhook signature mismatch

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Conventions

- **All timestamps:** ISO 8601 UTC (e.g., `2026-04-05T14:30:00.000Z`)
- **All IDs:** UUID v4
- **All monetary values:** Integer cents in API calls (e.g., `$45.00 = 4500`), NUMERIC(10,2) in database
- **Rate limiting:** 429 responses include `Retry-After` header with seconds to wait
- **Idempotency:** Webhook handlers use event_id for deduplication

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Endpoints

Health Check

```
GET /api/health
```

No authentication required. Used by deployment pipeline.

Response (200):

```
{
  "status": "ok",
  "db": "ok",
  "version": "abc1234"
}
```

Pharmacy Search

Search Medications (Autocomplete)

```
GET /api/pharmacy-search/medications?q={query}
```

Auth: Supabase JWT (clinic_user)

Parameters:

Param	Type	Required	Description

q	string	Yes	Medication name query (min 3 characters)

Response (200):

```
{
  "medications": [
    {
      "medication_name": "Semaglutide",
      "form": "Injectable",
      "dose": "0.5mg/0.5mL"
    }
  ]
}
```

Search Pharmacies

```
GET /api/pharmacy-search?medication={name}&form={form}&dose={dose}&state={state}
```

Auth: Supabase JWT (clinic_user)

Parameters:

Param	Type	Required	Description
medication	string	Yes	Medication name
form	string	No	Dosage form filter
dose	string	No	Dose filter
state	string	Yes	2-letter patient shipping state

Response (200):

```
{
  "results": [
    {
      "catalog_id": "uuid",
      "pharmacy_id": "uuid",
      "pharmacy_name": "Strive Pharmacy",
      "medication_name": "Semaglutide",
      "form": "Injectable",
      "dose": "0.5mg/0.5mL",
      "wholesale_price": 150.00,
      "average_turnaround_days": 5,
      "integration_tier": "TIER_4_FAX",
      "regulatory_status": "ACTIVE",
      "supports_real_time_status": false
    }
  ],
```

```
  "count": 1
}
```

Only pharmacies with an active license in the specified state are returned. BANNED pharmacies are excluded entirely.

Orders

Create Order

```
POST /api/orders
```

Auth: Supabase JWT (clinic_user)

Body:

```
{
  "patient_id": "uuid",
  "provider_id": "uuid",
  "catalog_id": "uuid",
  "pharmacy_id": "uuid",
  "retail_price": 300.00,
  "sig_text": "Take 0.5mg subcutaneous injection weekly"
}
```

Response (201):

```
{
  "order_id": "uuid",
  "status": "DRAFT"
}
```

Sign and Send Payment Link

```
POST /api/orders/{orderId}/sign-and-send
```

Auth: Supabase JWT (provider or clinic_admin)

Body:

```
{
  "signature_hash": "sha256-hex-string"
}
```

Transitions order from DRAFT to AWAITING_PAYMENT. Generates checkout JWT, dispatches SMS (if TWILIO_ENABLED=true), creates 3 SLA deadlines.

Response (200):

```
{
  "order_id": "uuid",
```

```
  "status": "AWAITING_PAYMENT",
  "checkout_url": "https://functional-medicine-
infrastructure.vercel.app/checkout/{jwt_token}"
}
```

Compliance Check

```
POST /api/orders/compliance-check
```

Auth: Supabase JWT (clinic_user)

Runs the 6 pre-dispatch compliance checks without actually sending.

Body:

```
{
  "order_id": "uuid"
}
```

Response (200):

```
{
  "all_passed": true,
  "checks": {
    "pharmacy_license": true,
    "provider_npi": true,
    "provider_signature": true,
    "price_validation": true,
    "stripe_connect_active": true,
    "dea_scheduling": true
  }
}
```

Checkout

Create Payment Intent

```
POST /api/checkout/payment-intent
```

Auth: Checkout JWT token (from URL)

Creates a Stripe PaymentIntent with Connect fee splitting.

Body:

```
{
  "order_id": "uuid",
  "email": "patient@example.com"
}
```

Response (200):

```
{
  "client_secret": "pi_xxx_secret_xxx",
  "amount": 30000,
  "currency": "usd"
}
```

Webhook Endpoints

All webhook endpoints return HTTP 200 regardless of internal processing result. Failures are handled via the Dead Letter Queue.

Stripe Webhooks

```
POST /api/webhooks/stripe
```

Auth: Stripe-Signature HMAC-SHA256 header

Events processed: payment_intent.succeeded, charge.dispute.created, transfer.failed, account.updated

Documo Webhooks

```
POST /api/webhooks/documo
POST /api/webhooks/documo/inbound
```

Auth: Documo webhook token verification

Events: fax.delivered, fax.failed, fax.received (inbound)

Twilio Webhooks

```
POST /api/webhooks/twilio
```

Auth: X-Twilio-Signature HMAC-SHA1

Events: SMS delivery status (delivered, failed, undelivered)

Pharmacy Webhooks

```
POST /api/webhooks/pharmacy/{pharmacySlug}
```

Auth: X-Pharmacy-Signature HMAC-SHA256 (per-pharmacy secret from Vault)

Events: order.acknowledged, order.compounding, order.rejected, order.shipped, order.delivered

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Cron Endpoints

All cron endpoints require `Authorization: Bearer {CRON_SECRET}` header.

Endpoint	Schedule	Purpose

POST /api/cron/sla-check	*/5 * * * *	SLA breach scanning
POST /api/cron/sla-refire	*/5 * * * *	Re-escalate stalled SLAs
POST /api/cron/payment-expiry	*/15 * * * *	Auto-expire unpaid orders
POST /api/cron/adapter-health-check	*/10 * * * *	Circuit breaker health
POST /api/cron/submission-reconciliation	*/30 * * * *	Detect stuck submissions
POST /api/cron/portal-status-poll	*/30 * * * *	Tier 2 status polling
POST /api/cron/fax-retry	*/5 * * * *	Fax retry with backoff
POST /api/cron/screenshot-cleanup	0 * * * *	Tier 2 screenshot cleanup
POST /api/cron/daily-digest	0 14 * * *	Daily ops digest to Slack

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Ops Endpoints

All ops endpoints require Supabase JWT with `app_role = 'ops_admin'`.

Endpoint	Method	Purpose
/api/ops/pipeline	GET	List all orders with filters
/api/ops/orders/{id}/detail	GET	Full order detail with history
/api/ops/orders/{id}/action	POST	Order actions (retry, reroute, cancel, add tracking)
/api/ops/sla	GET	List SLA deadlines with breach status
/api/ops/sla/acknowledge	POST	Acknowledge an SLA breach
/api/ops/adapters	GET	Pharmacy adapter health status
/api/ops/adapters/{pharmacyId}/action	POST	Adapter actions (disable, force fax, health check)
/api/ops/fax	GET	Inbound fax triage queue
/api/ops/fax/{faxId}/action	POST	Fax disposition (acknowledge, reject, query, unrelated)
/api/ops/catalog	GET	Catalog items with search
/api/ops/catalog/upload	POST	CSV catalog upload
/api/ops/catalog/item	POST	Manual catalog item entry
/api/ops/catalog/rollback	POST	Rollback catalog to previous version
/api/ops/catalog/sync/{pharmacyId}	POST	Trigger API catalog sync

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Rate Limits

Endpoint Category	Limit	Scope

/api/orders, /api/catalog, /api/pharmacy-search	100 req/min	Per clinic_id
/api/checkout/*	10 req/min	Per IP
/api/clinics/*/onboard	5 req/min	Per IP
/api/ops/adapters/*	20 req/min	Per ops_admin
/api/webhooks/*	No limit	N/A (rejecting causes data loss)
/api/cron/*	No limit	Server-only

All 429 responses include `Retry-After` header.

CompoundIQ API — All endpoints serverless. All state transitions atomic. All webhooks idempotent.